Literal weighted constructor invariant

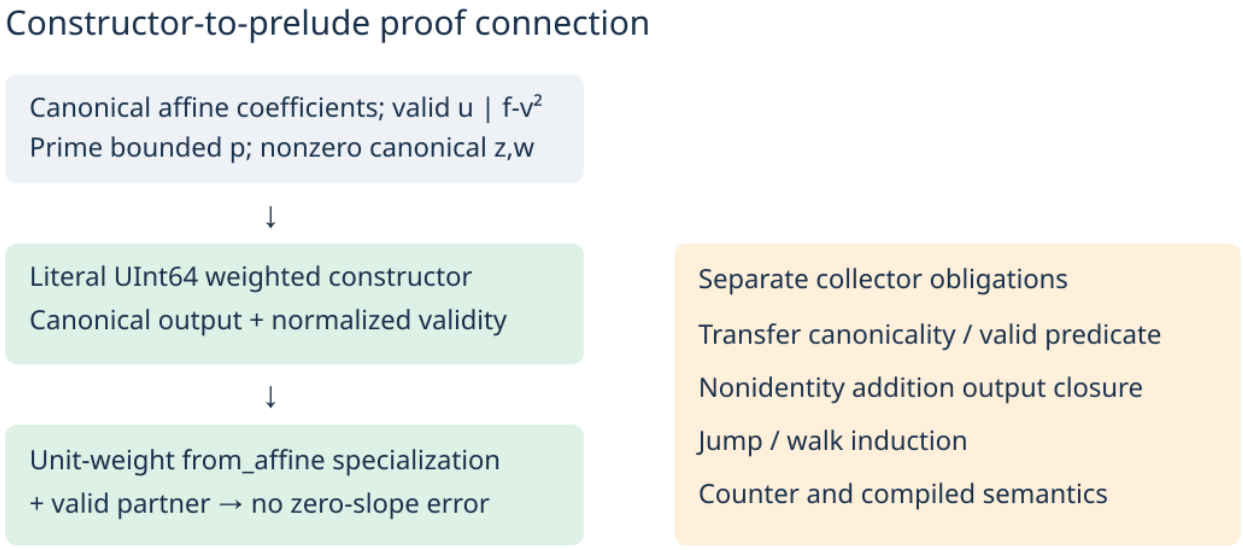
Private component proof packet; independent review pending

31 October 2026

Four Lean theorems compile: canonical output and field decoding of the literal UInt64 constructor; preservation of normalized Mumford validity and nonzero weights; the unit-weight degree-two from_affine specialization; and composition with the existing same-support zero-slope error exclusion. Separate controls and axiom readback pass.

Exact source connection

The constructor uses z^2, z^3, z^4, z^5 through reduced word products and returns $qz^2, rz^4, s(z^3w), t(z^5w), z, w$. The parenthesization $s(z^3w)$ matches the actual source; it differs from the earlier cross-scaling tree $(sz^3)w$. Existing bounded multiplication refinement supplies canonicality and decoding. Nonzero weights allow cancellation in the normalized Mumford divisibility predicate.



Green: compiled component statements. Amber: named next obligations. No cohort runs.

Assumptions and controls

The decoding theorem assumes positive $p < 2^{31}$ and canonical inputs. Validity preservation additionally assumes prime p, nonzero weights and affine support divisibility. The initialization/error composition uses $5 \le p < 2^{31}$. External Rust correspondence assumes Field.n=1, tables=None, normal execution and counter headroom. Shared expression subtrees are a value model, not an operation-count model.

A nonunit-weight control uses $p = 5$, $(q, r, s, t, z, w) = (0, 0, 0, 1, 2, 3)$ and $f = X^5 + X^2 + 1$. The full invariant holds; output $t = 1, z = 2, w = 3$ is checked. A zero-weight control confirms the constructor retains zero. Equal unit-weight inputs take the sameUDouble prelude branch.

Separate readback lists only propext and Quot.sound for decoding, and propext, Classical.choice and Quot.sound for the validity/composition statements. Both constructor compilation stages succeed. Raw source, commands and receipts are retained beside this PDF.

Collector boundary

The exact retained gate and collector sources match current repository HEAD (hash prefixes 5976e7f3 and 96246448). The collector checks initial affine validity before from_affine. This packet handles the degree-two constructor, conditional on canonical coefficients and the mathematical divisibility predicate. Connecting the native validity check and transfer output to those premises is separate. Identity addition passes through its operand; nonidentity addition closure is the next substantive dependency for jump/walk induction.

Next construction: generic branch output nonzero weights and normalized congruence, beginning with generic 1+2. No scientific trial or cost ratio is reported.

Evidence and prior work

See Constructor.lean, Literal.lean, Controls.lean, Readback.lean, sources.json and REPORT.md. Prior InvariantBranches and WordErrorExclusion results are reused. Literature assumptions follow Ozbay Gurler–Hisil, DOI 10.1007/s13389-024-00351-7, sections 2–4. This is an implementation connection, with no new addition-law claim.